

THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Privacy-Preserving Federated Learning for Distributed Skin Disease Classification

by
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Proposal Content Pages

- *The remainder of the template starting with 'Table of Contents' is offered as guidance only and is not intended to be prescriptive. You may adapt it to suit the specific requirements and conventions of your discipline and your project. The layout in this example is Introduction, Background and Literature Review, Project Plan but this may not suit your project! You can change these sections to whatever you need.*

- *The content pages that follow the preliminary pages are formatted using Microsoft Styles with embedded auto-numbering for chapters and headings. Page numbers have been incorporated to enable the automatic generation of the Table of Contents, as well as lists of tables and figures.*
- *If you include blank pages in your work, please mark the page with: "This page was intentionally left blank."*

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I acknowledge the use of generative AI tools in completing this assessment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content, and ensuring it adheres to academic integrity standards.

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- Explore your topic? Yes or No. Explain how (or explain why not).
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AI Model and date	Table 1. AI tool usage (tick all that apply)								
	Language Translation	Grammer / style / spelling	Topic exploration	Research question	Literature review	Content Creation visual (formatting)	Content Creation (e.g. code, text)	Feedback	Other (provide details)
ChatGPT 4o Weeks 5-6		✓	✓	✓				✓	
Midjourney 11/09/24						✓			

Abstract

Skin disease classification using deep learning has shown strong potential but training accurate models requires large, diverse datasets often distributed across healthcare institutions. Privacy regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) prevent centralised data aggregation. Prior work has shown that even sharing model updates rather than raw data can leak sensitive patient information. Federated learning offers a framework for collaborative training across distributed nodes without centralising data, but standard federated averaging lacks convergence guarantees. This project evaluates federated learning approaches for multi-class skin disease classification in a simulated distributed healthcare environment. An dataset of approximately 20,000 images covering 30 skin disease classes will be partitioned across multiple nodes. FedAvg and FedProx will be implemented and compared against a centralised training baseline, evaluating classification accuracy, communication efficiency, and model behaviour under data that is not independently and identically distributed (non-IID).

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3.0.2 Subsection as required 3

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List of Tables

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1. Introduction

Skin diseases are among the most common health conditions worldwide, and deep learning has demonstrated strong performance in automated skin disease classification from images. However, training models that generalise well requires large, diverse datasets that capture variation across patient populations, imaging conditions, and disease presentations. In practice, medical image data is distributed across hospitals, clinics, and research institutions, each holding relatively small local datasets. Privacy regulations including HIPAA and GDPR restrict the transfer and centralisation of patient data, creating a fundamental barrier to building robust classifiers through conventional centralised training.

Federated learning addresses this by enabling multiple institutions to collaboratively train a shared model without exchanging raw data. Each participant trains on its local dataset and shares only model updates with a central server, which aggregates them into a global model. McMahan et al. introduced Federated Averaging (FedAvg), which reduces communication costs by performing multiple local training epochs before aggregation [1]. However, FedAvg lacks convergence guarantees when data across participants is non-IID, a condition that is common in healthcare settings where different institutions may see different patient demographics and disease distributions. FedProx extends FedAvg by introducing a proximal term that constrains local model updates, providing convergence guarantees under heterogeneous conditions [2]. Furthermore, federated learning alone does not fully resolve privacy concerns — prior work has shown that shared model updates can leak sensitive information through inference attacks, and additional mechanisms such as secure aggregation and differential privacy may be needed [3].

This project evaluates federated learning for multi-class skin disease classification in a simulated distributed healthcare environment. A dataset of approximately 20,000 images covering 30 skin disease classes will be partitioned across multiple nodes. FedAvg and FedProx will be implemented and compared against a centralised training baseline, with evaluation focused on classification accuracy, communication efficiency, and model behaviour under non-IID data distributions. The remainder of this proposal is organised as follows. Chapter 2 provides background on federated learning, the challenges of statistical heterogeneity, and privacy considerations. Chapter 3 outlines the project plan, including methodology, milestones, and timeline.

2. Background and Literature review

You will need to review previous work in the field, which may include books and papers ("literature"), patents and commercial products ("prior art"), and earlier work in your Department. This information is usually (but not always) collected in a single chapter, whose title should preferably be more specific and interesting than the one above.

2.0.1 Subsections as required

You may want to use sub-headings here... you may not! Some sub-sections you might consider are:

- ***Background and theory*** *lay out the well-known theory and concepts (not cutting-edge research, could be found in a textbook) that must be understood to understand the literature review and your project proposal. If there is some nice background/theory that is complementary but not essential, consider pointing the reader to further sources (e.g. a textbook or paper) if they want a refresher.*
- ***Literature Review*** *outline the state of research, what has been done, and where the limitations are.*
- ***Literature/Knowledge Gaps*** *concisely summarize knowledge gaps. How you do this will depend on your project: you might wish to outline a broad gap, then narrow down toward the gap you are addressing in the project; you may wish to highlight several different gaps that need to be addressed and then narrow in on the one you will be addressing in your project.*

2.0.2 Subsections as required

Theory sections often contain equations like the equation for the Fourier series of the function $f(x)$:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right) \tag{2.1}$$

where a_0 is the average value of the function, and a_n and b_n are the coefficients for the nth harmonic ... etc.

3. Project Plan

3.0.1 Subsection as required

You may want to use sub-headings here... you may not!

Some sub-sections you might consider are:

- ***Project Aims*** *based on the literature gaps from the previous sections, what do you aim to do?*
- ***Methodology*** *What methods are going to be used and why. Are they faster, more efficient, all you have access to?*
- ***Milestones and Timelines*** *outline what will be done when to ultimately meet your aims. Are there any high-risk milestones you should identify which may require you to pivot your project in another direction?*
- ***Health and Safety and Risk Assessments***
- ***Ethics statement*** *if you have human/animal data*

3.0.2 Subsection as required

You will definitely need some figures and maybe some tables. As shown in Figure 3.1, insert your figure here.

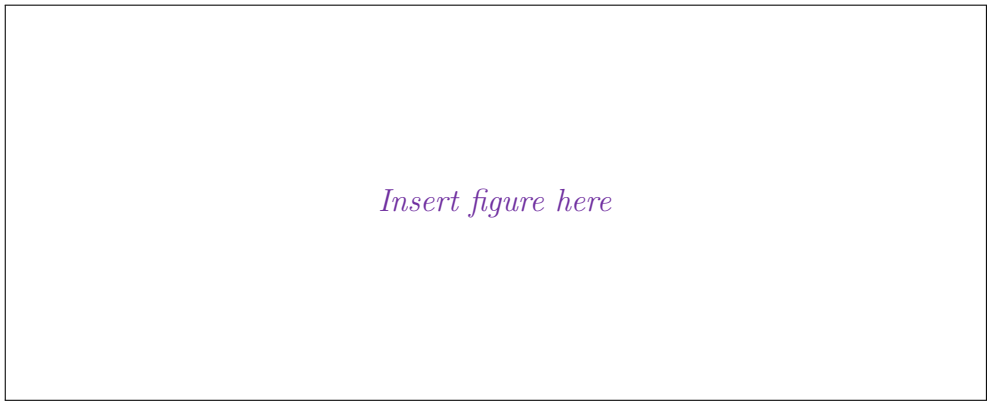


Figure 3.1: *Example figure caption. Replace this caption with a concise description of what the figure shows, including what the axes/labels represent and the key takeaway for the reader.*

Table 3.1 provides an example dataset showing how f, m, μ_2, and f_2 vary across the sampled conditions.

Table 3.1: *Example table caption. Replace this caption with what the variables represent (including units if applicable) and why this dataset is relevant to your proposal.*

$f(\%)$	m	μ_2	$f_2(\%)$
0.016	80.00	0.05400	4.874
0.031	56.57	0.07732	5.438
0.062	40.00	0.11103	6.125
0.125	28.28	0.16001	6.970
0.250	20.00	0.23175	8.020
0.500	14.14	0.33799	9.329
1.000	10.00	0.49789	10.967
2.000	7.07	0.74444	13.008
4.000	5.00	1.13919	15.525
8.000	3.54	1.81095	18.568
19.237	2.28	3.61958	23.174
37.180	1.64	7.28635	27.094
57.392	1.32	14.63631	29.813
74.316	1.16	29.35160	31.453
85.734	1.08	58.79364	32.360

Bibliography

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," Jan. 2023.

[2] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated Optimization in Heterogeneous Networks," Apr. 2020.

[3] G. A. Kaissis, M. R. Makowski, D. Rückert, and R. F. Braren, "Secure, privacy-preserving and federated machine learning in medical imaging," *Nature Machine Intelligence*, vol. 2, pp. 305–311, June 2020.